a grid-based puzzle game

COLOR SHIFT GRID

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INTRODUCTION

- Color Shift Grid is a grid-based puzzle game developed using Java and JavaFX.
- The game combines logical reasoning and strategic planning.
- Each move affects multiple cells simultaneously.
- Chain reactions create a dynamic gameplay experience.
- The project aims to provide challenging and replayable puzzle mechanics

MOTIVATION

- Puzzle games improve logical thinking and planning abilities
- Existing games often use repetitive mechanics

We wanted a game combining:

  - strategic gameplay
  - multiple objectives
  - replayability
  - dynamic interactions
  - The project also allowed application of OOP and UML principles

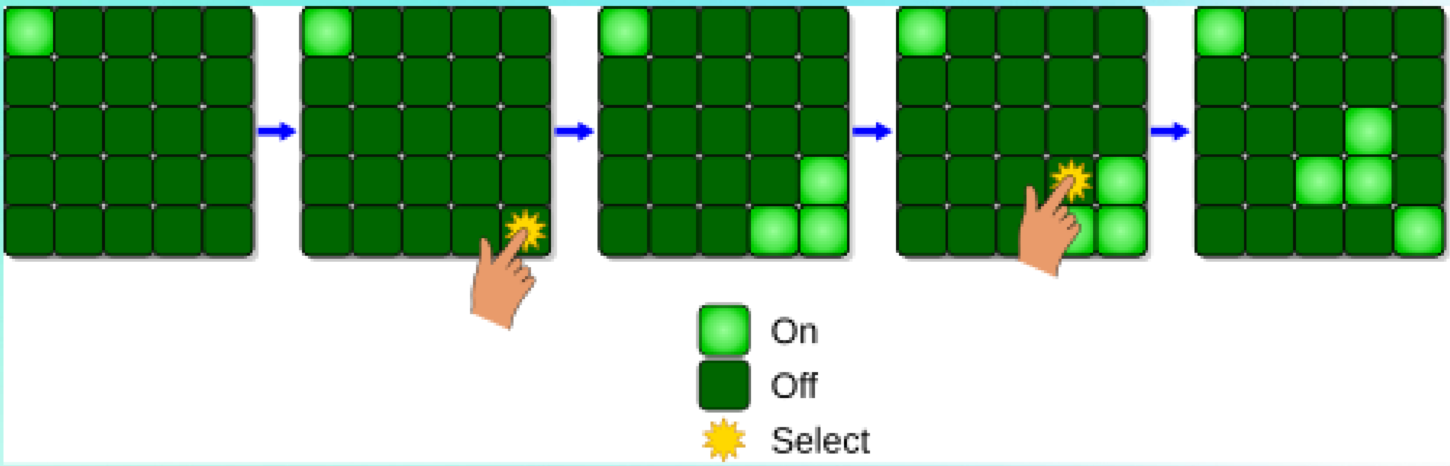
STATE OF THE ART

EXISTING APPLICATIONS

Color Shift Grid is inspired by grid-based puzzle games that require strategic thinking and pattern recognition. Several existing applications use similar mechanics involving color manipulation and neighbor interactions.

1. Lights Out

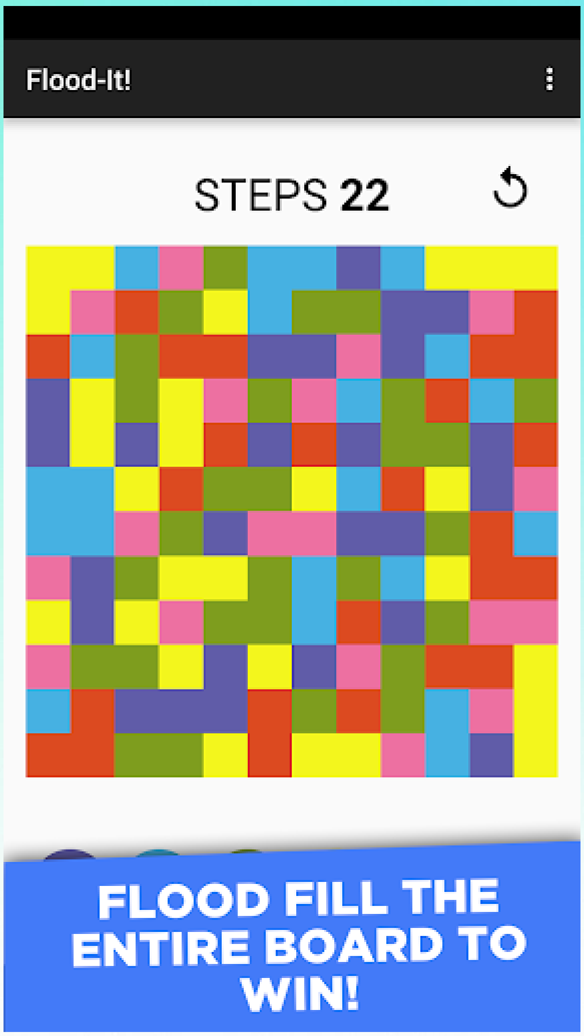
- Grid-based puzzle where clicking a cell changes its state and adjacent cells.
- Objective: turn all lights off in minimum moves
- Similarity:
  - neighbor-based updates
  - chain reactions
- Difference:
- Color Shift Grid introduces multiple colors and game modes

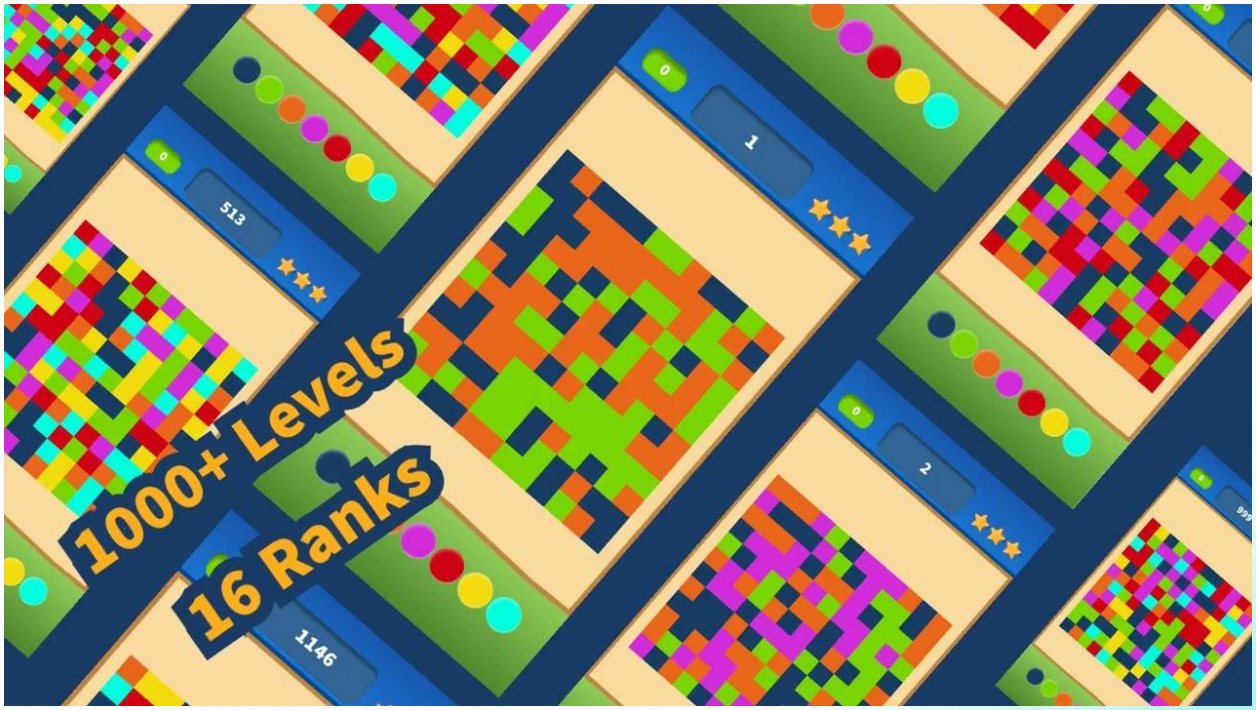

On
Off
Select


LIGHTS
OUT
ON/OFF
START
SOUND
HELP
SELECT
TIGER

2. Flood-It

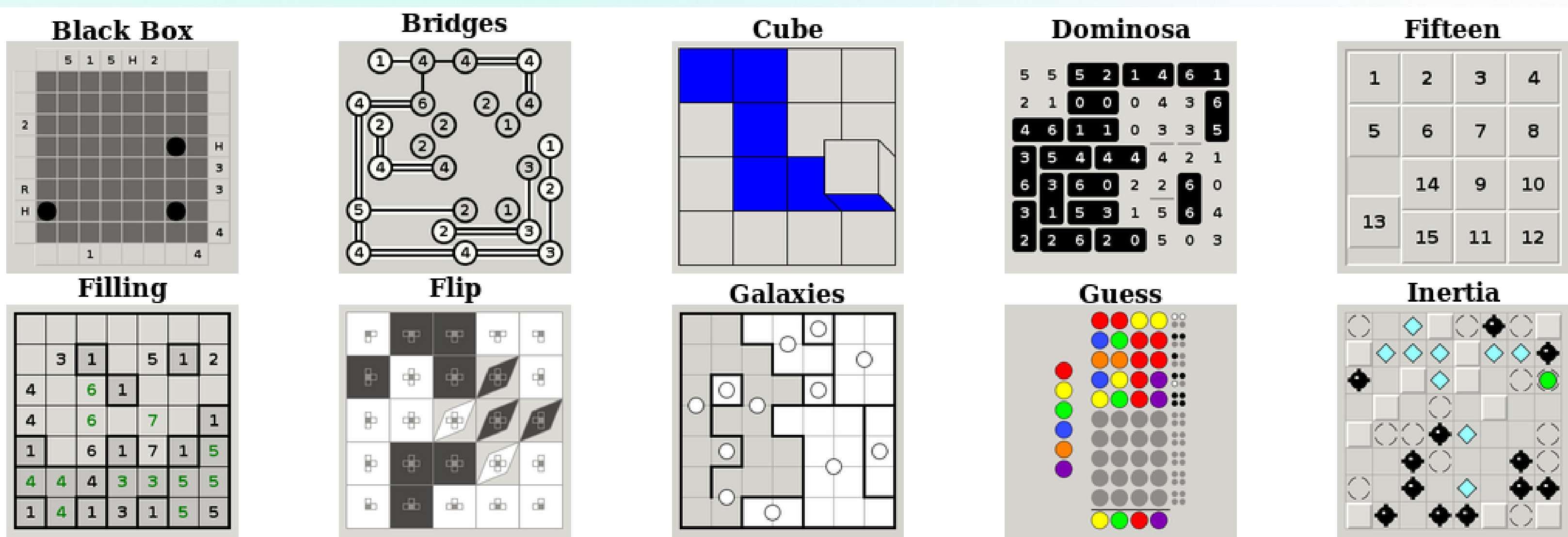
- Puzzle game where the player fills the board using one color within a move limit
- Uses strategic color planning and optimization
- Similarity:
  - color-based gameplay
  - move constraints
- Difference:
  - Color Shift Grid modifies local cells and neighbors instead of flood filling


Flood-It!
STEPS 22
FLOOD FILL THE ENTIRE BOARD TO WIN!


1000+ Levels
16 Ranks

3. Logic Grid Puzzle Collections

- Puzzle collections such as Simon Tatham puzzles include strategy and grid reasoning mechanics
- Focus on planning and problem solving



PROJECT SPECIFICATIONS

Purpose of the Application

- interactive grid-based puzzle game
- develops logical thinking and strategic planning
- each move affects multiple elements
- creates chain reactions across the grid

Functional Requirements

The system must provide:

- 2D grid (e.g. 3x3) with colored cells
- user interaction through cell clicks
- color transformation (cell + neighbors)
- game modes: classic, challenge and pattern
- undo and restart functionality

Game Rules

- each cell stores a color as an integer (0-3)
- colors follow a cyclic order:

Red – Yellow – Green – Blue – Red

- a move updates: the selected cell and the direct neighbors (N, S, E, W)
- update rule:

new = (old+1) mod 4

- win condition depends on the selected game mode

Game modes

- Classic Mode
  - no move limit
  - undo
  - restart
  - progress
- Challenge Mode
  - move limit
  - limited undo
  - restart
  - progress
- Pattern Mode
  - target pattern
  - undo
  - restart
  - progress

Level Generation

- start from a solved grid
- apply a sequence of valid moves
- resulting grid becomes the level
- ensures every level is solvable

Constraints

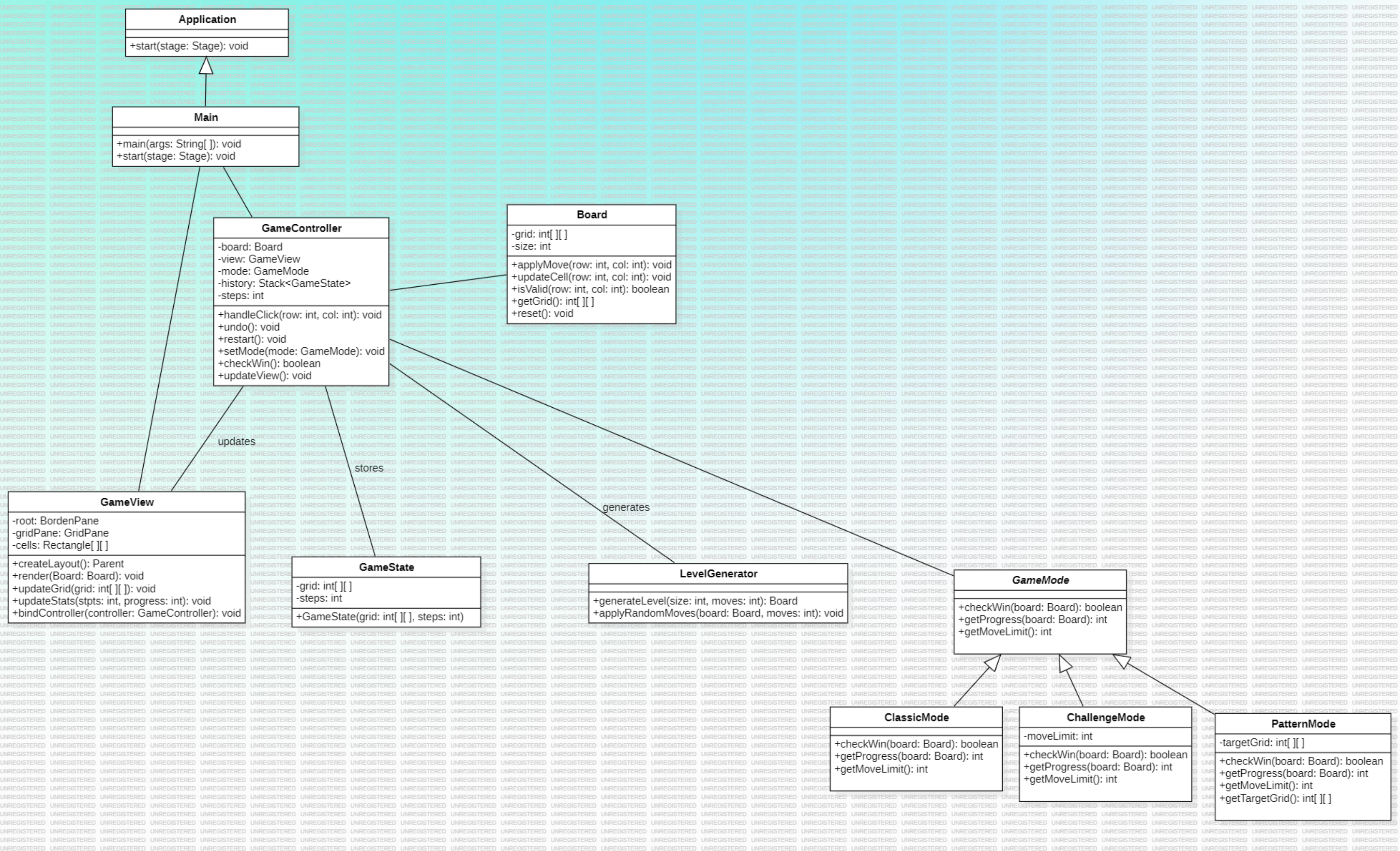
- grid size is fixed (e.g. 3x3)
- only valid positions are affected by moves
- game rules are deterministic
- constraints vary depending on game mode

UML OVERVIEW

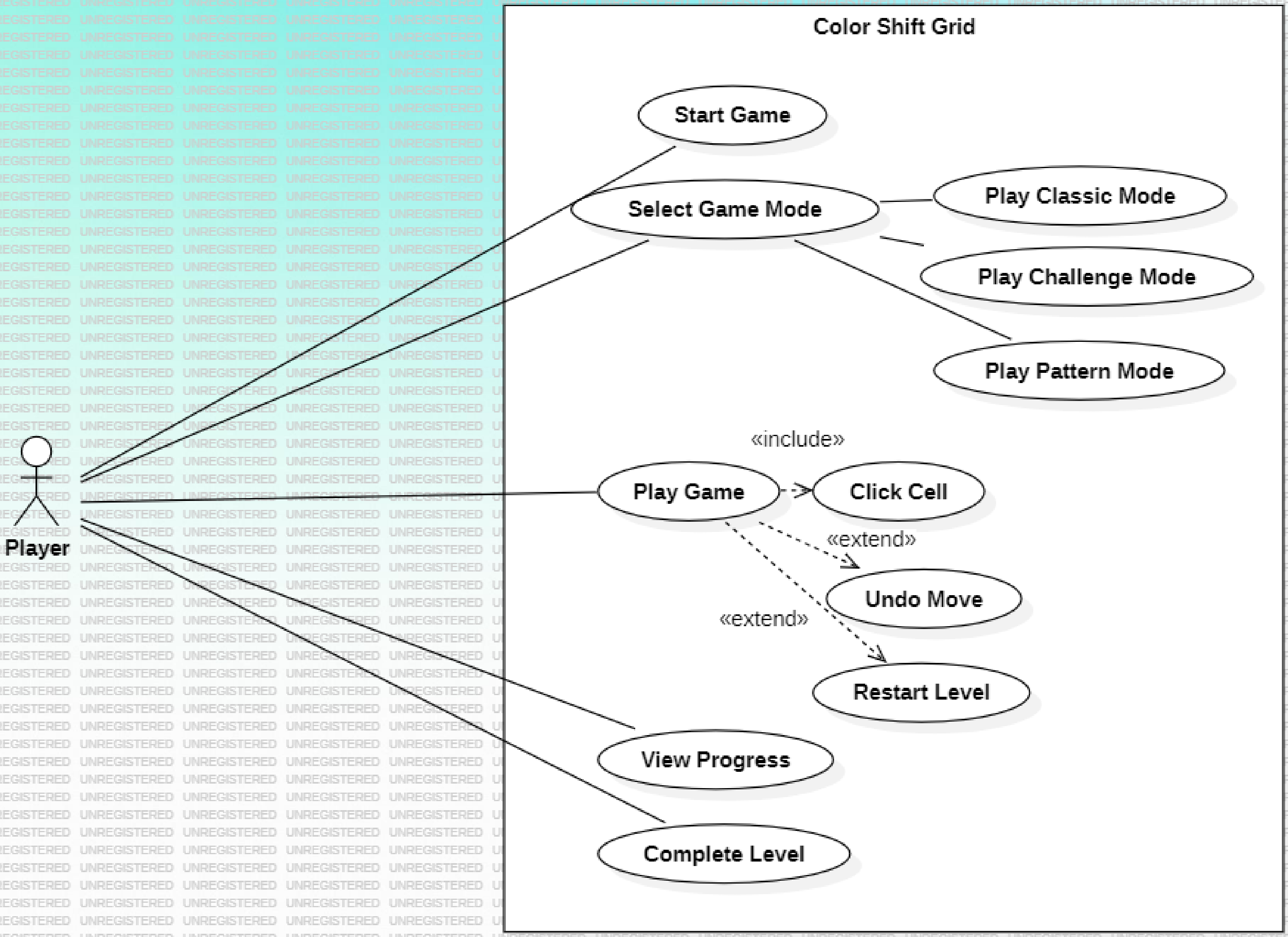
System Modeling (UML)

- Class Diagram
- Use Case Diagram
- State Diagram

CLASS DIAGRAM



USE CASE DIAGRAM



STATE DIAGRAM